

Problem 1.

Convert each decimal number to hexadecimal:

26 = _____

60 = _____

175 = _____

255 = _____

Problem 2.

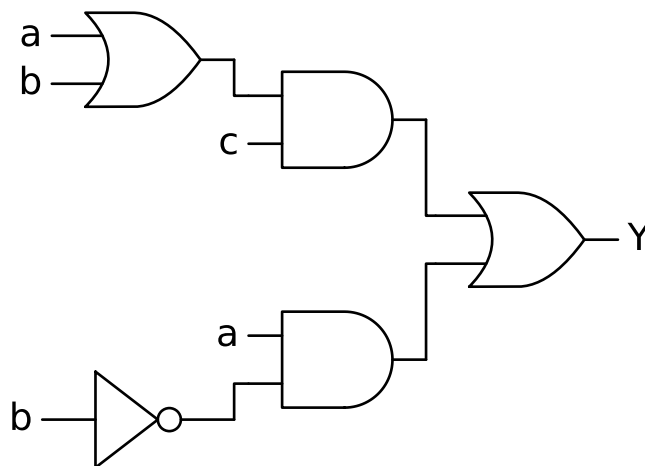
Convert each hexadecimal number to decimal:

1A₁₆ = _____

2F₁₆ = _____

B4₁₆ = _____

FF₁₆ = _____

Problem 3.

Write the Boolean expression for output Y , then draw its truth table.

Problem 4.

Convert each binary number to hexadecimal:

$$11010_2 = \underline{\hspace{2cm}}$$

$$101111_2 = \underline{\hspace{2cm}}$$

$$10110100_2 = \underline{\hspace{2cm}}$$

$$1111_2 = \underline{\hspace{2cm}}$$

Problem 5.

Convert each hexadecimal number to binary:

$$3C_{16} = \underline{\hspace{2cm}}$$

$$A7_{16} = \underline{\hspace{2cm}}$$

$$1E_{16} = \underline{\hspace{2cm}}$$

$$FF_{16} = \underline{\hspace{2cm}}$$

Problem 6.

Every device on a network has an *IP address*: four numbers separated by dots, each stored in one byte (so 0 to 255). Convert each part of these addresses to 8-bit binary.

$$192.168.1.1 = \underline{\hspace{1cm}} \cdot \underline{\hspace{1cm}} \cdot \underline{\hspace{1cm}} \cdot \underline{\hspace{1cm}}$$

$$10.0.0.255 = \underline{\hspace{1cm}} \cdot \underline{\hspace{1cm}} \cdot \underline{\hspace{1cm}} \cdot \underline{\hspace{1cm}}$$

$$8.8.8.8 = \underline{\hspace{1cm}} \cdot \underline{\hspace{1cm}} \cdot \underline{\hspace{1cm}} \cdot \underline{\hspace{1cm}}$$

Which one of these could *not* be a real IP address, and why?

192.168.0.1 10.0.256.4 172.16.0.255

Fun fact: 8.8.8.8 is one of Google's real public addresses.

Problem 7.

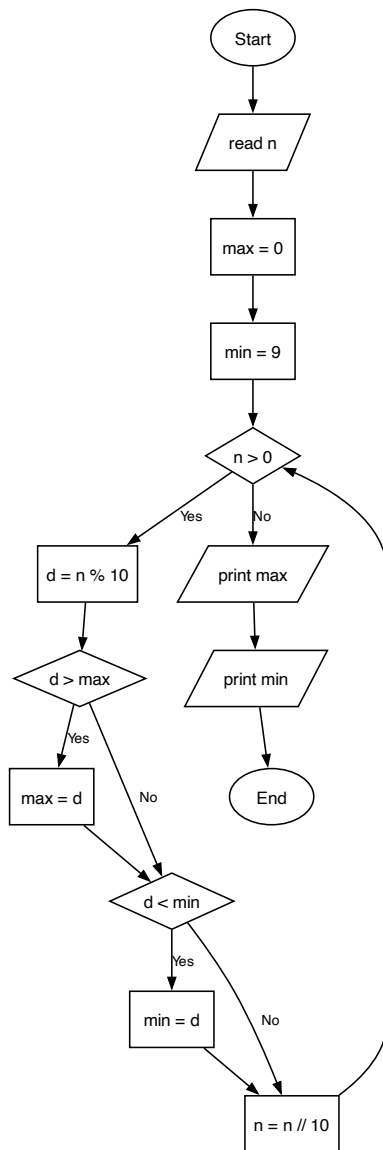
A black-and-white picture is a grid of *pixels*: each pixel is one bit — 1 means a filled (black) square, 0 means blank. Each row of 8 pixels is one byte, given below in hexadecimal. Convert each byte to 8-bit binary, then shade a square wherever there is a 1.

66 ₁₆ =	_____	<table><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>								
FF ₁₆ =	_____	<table><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>								
FF ₁₆ =	_____	<table><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>								
FF ₁₆ =	_____	<table><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>								
7E ₁₆ =	_____	<table><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>								
3C ₁₆ =	_____	<table><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>								
18 ₁₆ =	_____	<table><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>								
00 ₁₆ =	_____	<table><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>								

What picture appears? The whole image is just 8 bytes — 64 ones and zeros. Roughly how many bytes would a picture twice as wide and twice as tall (16×16) need?

Problem 8.

Trace this algorithm for $n = 5184$, and complete the trace table.

[illegible]